

DSA0502 – QUERY PROCESSING FOR DATA SCIENCE

LAB PROGRAMS (15-16)

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15. Write a Pandas program to keep only the rows with at least two NaN values in a given DataFrame.

Aim

To filter and display only those rows that contain at least two missing (NaN) values using Pandas.

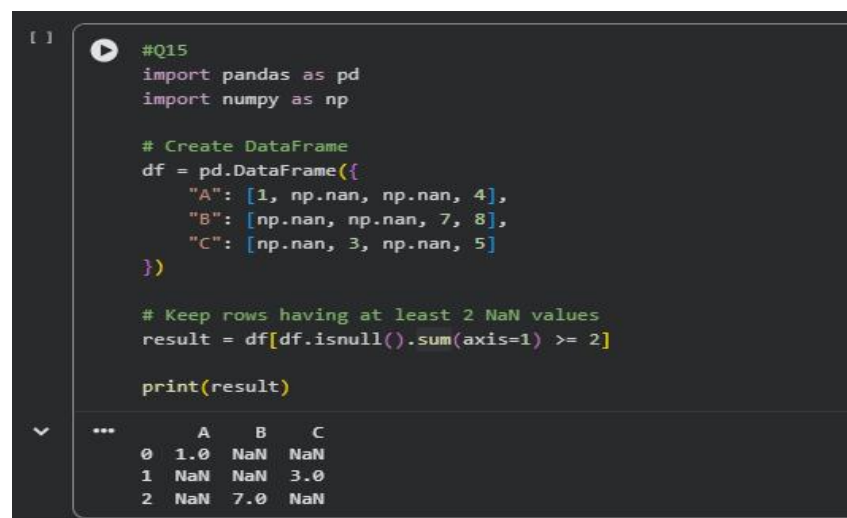
Code

```
import pandas as pd
import numpy as np

# Create DataFrame
df = pd.DataFrame({
    "A": [1, np.nan, np.nan, 4],
    "B": [np.nan, np.nan, 7, 8],
    "C": [np.nan, 3, np.nan, 5]
})

# Keep rows having at least 2 NaN values
result = df[df.isnull().sum(axis=1) >= 2]
print(result)
```

OUTPUT



```
[ ] #Q15
import pandas as pd
import numpy as np

# Create DataFrame
df = pd.DataFrame({
    "A": [1, np.nan, np.nan, 4],
    "B": [np.nan, np.nan, 7, 8],
    "C": [np.nan, 3, np.nan, 5]
})

# Keep rows having at least 2 NaN values
result = df[df.isnull().sum(axis=1) >= 2]

print(result)
```

	A	B	C
0	1.0	NaN	NaN
1	NaN	NaN	3.0
2	NaN	7.0	NaN

16. Write a Pandas program to split the given DataFrame into groups based on the school code. Also check the type of the GroupBy object.

Aim

To group a DataFrame based on the **School_Code** column using the `groupby()` method and determine the type of the resulting GroupBy object.

Code

```
import pandas as pd

# Create DataFrame
df = pd.DataFrame({
    "Name": ["A", "B", "C", "D", "E"],
    "School_Code": [101, 102, 101, 103, 102],
    "Marks": [80, 75, 90, 85, 70]
})

# Group by School_Code
group = df.groupby("School_Code")

# Display groups
for code, data in group:
    print("\nSchool Code:", code)
    print(data)

# Check type
print("\nType of GroupBy Object:")
print(type(group))
```

OUTPUT

[]



```
#Q16
import pandas as pd

# Create DataFrame
df = pd.DataFrame({
    "Name": ["A", "B", "C", "D", "E"],
    "School_Code": [101, 102, 101, 103, 102],
    "Marks": [80, 75, 90, 85, 70]
})

# Group by School_Code
group = df.groupby("School_Code")

# Display groups
for code, data in group:
    print("\nSchool Code:", code)
    print(data)

# Check type
print("\nType of GroupBy Object:")
print(type(group))
```



...

```
School Code: 101
  Name  School_Code  Marks
0    A           101     80
2    C           101     90
```

```
School Code: 102
  Name  School_Code  Marks
1    B           102     75
4    E           102     70
```

```
School Code: 103
  Name  School_Code  Marks
3    D           103     85
```

```
Type of GroupBy Object:
<class 'pandas.core.groupby.generic.DataFrameGroupBy'>
```